



Practical-4

TF-IDF

Data frame

```
import os
import pandas as pd
from sklearn.feature_extraction.text
import TfidfVectorizer
```

```
def load_books(directory):
    documents = []
    filenames = []
```

```
    for filename in os.listdir(directory):
        with open(os.path.join(directory, filename), 'r', encoding='utf-8') as file:
```

```
            documents.append(file.read())
            filenames.append(filename)
```

```
    return documents, filenames
```

```
books_dir = '/path/to/gutenberg-books'
documents, filenames = load_books(books_dir)
```

```
vectorizer = TfidfVectorizer(stop_words='english', max_df=0.95, min_df=2)
```

```
tfidf_df = pd.DataFrame(tfidf_matrix.toarray(),
                        index=filenames, columns=terms)
```

```
print(tfidf_df)
```